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Actuation mechanism for coupling device

Abstract

An actuation mechanism for use with a coupling device. The actuation mechanism includes an assembly having a hub, a latch member, a translating member, and an indexer. The assembly includes a carrier supported on the hub that engages the indexer. The carrier moves axially relative to the hub. A return member biases the indexer to a non-deployed position, with the translating member acting against the return member to move the indexer to the deployed position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage of International Application No. PCT/US2022/052220, filed Dec. 8, 2022, and claims the benefit of U.S. Provisional Application No. 63/287,320, filed Dec. 8, 2021. The disclosure of the above applications is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention generally relates to an actuation mechanism and, more specifically, an actuator mechanism for use with a coupling device such as a one-way clutch or brake.

2. Description of Related Art

(2) One-way clutches, or brakes, may include a strut or coupling member movable to an engaged/locked position. An actuator or actuation mechanism may control the strut or coupling member movement and move it to the engaged/locked position.

SUMMARY OF THE INVENTION

(3) An assembly including a hub. The hub includes a hub having a rotational axis. The assembly may also include a latch member, the latch member is rotatably and axially fixed to the hub, a translating member including an annular translator attached to an outer circumferential surface thereof, the translating member supported on the latch member, the translating member is rotatably fixed and axially movable relative to the rotational axis and an indexer supported on the hub, the indexer rotatably and axially movable relative to the rotational axis. A carrier supported on the hub engages the indexer. The carrier is rotatably fixed and axially movable for axial motion with no rotation relative to the hub. A return member moves the indexer to a non-deployed position, with the translating member acting against the return member to move the indexer to the deployed position.

(4) Further areas of applicability of the present invention will become apparent from the detailed description provided. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1 is a schematic, cross-sectional perspective view of an actuation mechanism for use with a coupling device viewed from the right.

(3) FIG. 2 is a schematic, cross-sectional perspective view of the actuation mechanism of FIG. 1 viewed from the left.

(4) FIG. 3 is a schematic, partially exploded, perspective view of the actuation mechanism of FIG. 1 viewed from the left.

(5) FIG. 3A is an enlarged view of the latch or stop member of the actuation mechanism of FIG. 3.

(6) FIG. 3B is an enlarged view of the translating member of the actuation mechanism of FIG. 3.

(7) FIG. 3C is an enlarged view of the indexer of the actuation mechanism of FIG. 3.

(8) FIG. 4 is a schematic, partially exploded, perspective view of the actuation mechanism of FIG. 1 viewed from the right.

(9) FIG. 4A is an enlarged view of the latch or stop member of the actuation mechanism of FIG. 4.

(10) FIG. 4B is an enlarged view of the translating member of the actuation mechanism of FIG. 4.

(11) FIG. 4C is an enlarged view of the indexer of the actuation mechanism of FIG. 4.

(12) FIG. 5 is a schematic, sectional, perspective view of the actuation mechanism of FIG. 1.

- (13) FIG. 6 is a schematic, sectional, perspective view, with portions removed illustrating the actuation mechanism of FIG. 1 in a non-deployed position.
- (14) FIG. 7 is a schematic, sectional, perspective view, with portions removed illustrating the actuation mechanism of FIG. 1 moving from a non-deployed position.
- (15) FIG. 8 is a schematic, sectional, perspective view, with portions removed illustrating the actuation mechanism of FIG. 1 in a deployed position.
- (16) FIG. 9 is a schematic, sectional, perspective view, with portions removed illustrating the actuation mechanism of FIG. 1 moving from a deployed position.
- (17) FIG. 10 is a perspective view of an additional embodiment of an actuation mechanism.
- (18) FIG. 11 is a cross-sectional perspective view of the actuation mechanism of FIG. 10.
- (19) FIG. 12 is a perspective view of another embodiment of an actuation mechanism.
- (20) FIG. 13 is a cross-sectional perspective view of the actuation mechanism of FIG. 12.
- (21) FIG. 14 is a perspective view of a further embodiment of an actuation mechanism.
- (22) FIG. 15 is a cross-sectional perspective view of the actuation mechanism of FIG. 14.
- (23) FIG. 16 is a schematic, cross-sectional view of the actuation mechanism for use with a coupling device illustrating stator operation.
- (24) FIG. 17 is a schematic, exploded, partial view of another embodiment of a stator and a translator assembly used with the actuation mechanism.
- (25) FIG. 18 is a schematic, partial side view of another embodiment of an actuation mechanism for use with a coupling device.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (26) The following description of the preferred embodiment(s) is merely exemplary in nature and is not intended to limit the invention, its application, or its uses.
- (27) Examples of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of the components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.
- (28) FIGS. 1-4 illustrate a first example of an actuation mechanism, generally seen at 10, and a coupling device, generally seen at 12. One example of the coupling device 12 is a controllable clutch including an annular member, sometimes referred to as a pocket plate 14, having a recess or pocket 16 in a face or side surface 15 of the pocket plate 14 As used herein, the term “clutch” should be interpreted to include clutches or brakes wherein one plate is drivably connected to a torque delivery element of a transmission, and the other plate is drivably connected to another torque delivery element or is anchored and held stationary to a housing. The terms “coupling,” “clutch,” and “brake” may be used interchangeably.
- (29) A drive member or biasing member, shown herein as a spring 20, outwardly biases a locking element, shown as a strut 18, pivotally mounted in the recess or pocket 16 in the pocket plate. The biasing member or spring 20 extends through an aperture 22 in the pocket plate 14. One end of the biasing member or spring 20 engages an end of the strut 18, and the opposite end engages a spring plate 24. The actuation mechanism 10 controls the movement of the strut 18. The actuation mechanism 10 moves the spring plate 24 and biasing member or spring 20 linearly toward the pocket plate 14 from a clutch “off” position to a clutch “on” position wherein in the clutch “on” position, shown in FIG. 1, the strut 18 moves outwardly past the recess or pocket 16. The clutch “off” position and clutch “on” position may also be referred to as a non-deployed position, clutch “off” and a deployed position, clutch “on.” For example, the strut 18, located in the pocket plate 14 moves between a between a non-deployed position—the strut 18 located in the pocket 16 and a deployed position—the strut 18 extending outwardly from the pocket 16 and beyond or past the face or side surface 15 of the pocket plate 14.

(30) A return member, shown as a spring **26**, extends between a bore **28** in the pocket plate **14** and the spring plate **24**. The return member or spring **26** biases the pocket plate **14** and the spring plate **24** apart by moving the spring plate **24** away from the pocket plate **14**.

(31) The coupling device **12** may include a cover or selector plate **30** positioned adjacent the face or side surface **15** of the pocket plate **14**.

(32) Instead of the drive or biasing member being a coiled spring **20**, the spring plate **24** can be operatively connected to other types of strut actuators, such as pins, which linearly move the struts **18** outwardly of the recess or pocket **16** and past the face or side surface **15** of the pocket plate **14**.

(33) One example of the actuation mechanism **10** includes an elongated, cylindrical hub **32**, extending along and rotating about a longitudinal axis **34**, a spring plate carrier **36**, an indexer **38**, a latch or stop member **40**, a stator **42**, a translator **44**, and a translating member **46**. Depending upon the relative positions of the indexer **38** and latch or stop member **40**, the actuation mechanism **10** moves between an initial or retracted position and a subsequent or extended position. The return member or spring **26** operates to bias or urge the actuation mechanism **10** to the initial or retracted position.

(34) The hub **32** includes a plurality of elongated splines **48** extending longitudinally along an outer circumferential surface **50** of the hub **32**. The splines **48** are a plurality of radially extending projections **48a**, external splines, separated by grooves **48b**, internal splines, formed longitudinally around the outer circumferential surface **50** of the hub **32**. The splines **48** fit into corresponding grooves, internal splines, and projections, external splines in a mating part.

(35) The spring plate carrier **36** includes a cylindrical body **52** and a radially outwardly extending flange **54**. The cylindrical body **52** of the spring plate carrier **36** has radially inwardly extending splines **56**. The splines **56** mate with the grooves **48b** of the splines **48** on the outer circumferential surface **50** of the hub **32**. The cylindrical body **52** is positioned and sliding longitudinally on the outer circumferential surface **50** of the hub **32**. The corresponding splines **48**, **56** prevent relative rotation between the spring plate carrier **36** and the hub **32**. The splined connection between the spring plate carrier **36** and hub **32** permits reciprocal axial movement of the spring plate carrier **36** along the hub **32** in the direction of the longitudinal axis **34**. The radially outwardly extending flange **54** of the spring plate carrier **36** engages the spring plate **24**, wherein longitudinal movement of the spring plate carrier **36** correspondingly moves the spring plate **24**. In an alternative embodiment, the spring plate **24** and the carrier **36** can be an integral or unitary member.

(36) The indexer **38** has a cylindrical body **58** having an inner circumferential surface **60** and an outer circumferential surface **62**. The cylindrical body **58** of the indexer **38** is positioned over the cylindrical body **52** of the spring plate carrier **36**; specifically, the inner circumferential surface **60** of the cylindrical body **58** of the indexer **38** is located adjacent to an outer circumferential surface **64** of the cylindrical body **52** of the spring plate carrier **36**. One end **66** of the indexer **38** engages the radially outwardly extending flange **54** of the spring plate carrier **36**. The opposite end **68** of the indexer **38** includes a plurality of longitudinally extending first or outer cam members **70** disposed circumferentially about the opposite end **68** in a saw-tooth configuration—each outer cam member **70** having a cam surface or ramp **72** and a cam displacement surface **74**. The first or outer cam members **70** are adjacent to the outer circumferential surface **62** of the indexer **38**. The indexer **38** moves through incremental positions and cooperates with other components of the actuation mechanism **10** to locate or place the actuation mechanism in either an “on” or “off” position.

(37) The inner circumferential surface **60** of the indexer **38** includes a plurality of splines **76** formed of longitudinally extending grooves/guides **78** and projections or fingers **80**. The projections or fingers **80** extending to the end **68** of the indexer **38**. The projections or fingers **80** forming second or inner longitudinally extending cam members **82** disposed circumferentially about the opposite end **68**. The second cam members **82** are adjacent to the inner circumferential surface **60** of the indexer **38**. Each finger **80** includes a cam surface or ramp **84**, extending at the same angle and in common with the cam surface or ramp **72** of the outer cam members **70**. The

groove or guide **78** terminates or ends the cam surface or ramp **84** of the projection or finger **80** before the cam surface or ramp **84** reaches the vertex **86** of a first or outer cam member **70**, the vertex being the intersection of the cam surface or ramp **72** and the cam displacement surface **74**.

(38) The indexer **38** rotates about the hub **32** and the spring plate carrier **36** and slides axially to both. As shown in FIGS. 5-9, the indexer **38** rotates circumferentially in the direction of the arrow **152**. The indexer **38** also moves reciprocally, in an axial direction, in the direction of arrows **150**, **160**, along the longitudinal axis **34** of the hub **32** of the actuator mechanism **10**.

(39) The translating member **46** includes a plurality of longitudinally extending first or outer cam members **94** disposed circumferentially about one end **95** of the translating member **46** in a saw-tooth configuration—each cam member **94** having a cam surface or ramp **96** and a cam displacement surface **98**. The first or outer cam members **94** are adjacent to the outer circumferential surface **132** of the translating member **46**.

(40) An inner circumferential surface **134** of the translating member **46** includes a plurality of splines **88** formed of longitudinally extending grooves or guides **90** and projections or fingers **92**. The projections or fingers **92** extend longitudinally from the end **95** of the translating member **46**. The ends of the projections or fingers **92** form second or inner cam members **136**, which are adjacent to the first or outer cam members **94**. The second or inner cam members **136** include cam surfaces or ramps **138** and cam displacement surfaces **140**. Similar to the respective cam surfaces or ramps **72**, **84** of the outer and inner cam members **70**, **82**, the respective cam surfaces or ramps **138** of the inner cam members **136** of the translating member **46** extend at the same angle and in common with the cam surface or ramps **96** of the first or outer cam members **94**.

(41) While at the same angle and in common, the respective cam surfaces or ramps of both sets of outer cam members **70**, **82** and **94**, **136** could have different angles and lie at different circumferential positions, for example, spaced apart.

(42) The first or outer members **94**, and corresponding cam surfaces or ramps **96** and cam displacement surfaces **98**, of the translating member **46** cooperate and work with the corresponding or complementary first or outer cam members **70**, and corresponding cam surfaces or ramps **72** and cam displacement surfaces **74**, of the indexer **38**. While the second or inner cam members **136**, and corresponding cam surfaces or ramps **138** and cam displacement surfaces **140**, of the translating member **46** cooperate and work with the corresponding or complementary second or inner cam members **82**, and corresponding cam surfaces or ramps **84** and cam displacement surfaces **85**, of the indexer **38**.

(43) The latch or stop member **40** is fixed to the hub **32**. As shown in FIG. 1, a longitudinal end **100** of the latch or stop member **40** engages a shoulder **102** on the hub **32**. The opposite longitudinal end **104** of the latch or stop member **40** is held stationary on the hub **32** by a snap ring **106** in a circumferential groove **108** in the hub **32**. An inner circumferential surface **110** of the latch or stop member **40** includes splines **112** engaging the splines **48** of the hub **32** to prevent rotation of the latch or stop member **40** with respect to the hub **32**. An outer circumferential surface **114** of the latch or stop member **40** includes a plurality of longitudinally extending splines **116** formed of longitudinally extending grooves or guides **118** and projections or fingers **120**. The end **104**, adjacent to the snap ring **106**, has an annular portion **122** forming an end wall or surface **124** of the grooves or guides **118**.

(44) The latch or stop member **40** includes a plurality of longitudinally extending cam members **126** disposed circumferentially at the ends of the projections or fingers **120** near the end **100** of the latch or stop member **40** contacting the shoulder **102** of the hub **32**. Each cam member **126** of the latch or stop member **40** includes a cam surface or ramp **128** and a cam displacement surface **130**. The cam members **126** have a saw-tooth configuration.

(45) The cam surface or ramp **128** is complementary to and has the same angle as the cam surface or ramp **84** of the second cam **82** of the indexer **38**. The cam displacement surface **130** is complementary to and has the same angle as the cam displacement surface **85** of the indexer **38**.

(46) The grooves or guides **118** on the outer circumferential surface **114** of the latch or stop member **40** have the same circumferential width as the circumferential width of each longitudinally extending projection or finger **80** of the indexer **38**. As illustrated in FIG. **4**, the longitudinally extending projections or fingers **80** of the indexer **38** slide axially in the grooves or guides **118** of the latch or stop member **40**.

(47) In addition to cooperating with the inner cam members **136** of the translating member **46**, the second or inner cam members **82** of the indexer **38** also cooperate with the cam members **126** of the latch or stop member **40**.

(48) The stator **42** provides a magnetic field that acts on the translator **44**, in the present example, a steel ring. Electric current applied to the stator **42** creates a magnetic field that acts on and moves the translator **44**. Because the translator **44** is attached to the translating member **46**, secured to the outer circumferential surface **132** thereof, movement of the translator **44**, caused by the stator **42**, results in movement of the translator **44** and translating member **46** combination. The projections or fingers **92** on the inner circumferential surface **132** of the translating member **46** engage the grooves or guides **118** of the latch or stop member **40**. The translating member **46** and translator **44** move axially along the latch or stop member **40** in the direction of the longitudinal axis **34** but do not rotate about the latch or stop member **40**. The translatory or sliding motion of the translating member **46** on the latch or stop member **40** is similar to that of a plunger moving back and forth in a reciprocal motion.

(49) As the translating member **46** moves longitudinally in the direction of the longitudinal axis **34**, the cam surfaces or ramps **96** and cam displacement surfaces **98** of the cam members **94** of the translating member **46** interact with corresponding cam surfaces or ramps **72** and cam displacement surfaces **74** of the outer cam members **70** of the indexer **38** to both axially move and rotate the indexer **38**.

(50) The actuation mechanism **10** moves the coupling device **12** between an “off” position and an “on” position. In the “on” position, the actuation mechanism **10** extends and moves the spring plate carrier **36** and spring plate **24** combination closer to the pocket plate **14**, wherein the springs **20** act on the struts **18**. In the “off” position, the actuation mechanism **10** retracts, moves to the initial position, enabling the spring plate carrier **36** and spring plate **24** combination to be spaced further from the pocket plate **14**. The “on” and “off” positions of the coupling device **12** can be established by the location of, or the relative engagement of, the indexer **38** and the latch or stop member **40**.

(51) FIGS. **5-6** illustrate the actuation mechanism **10** in a retracted position placing the coupling device **12** in the “off” position, with the stator **42** and translator **44** removed for clarity in FIGS. **6-9**. The retracted position occurs when the cam surface or ramp **84** and projection or finger **80** of the indexer **38** are positioned in the grooves or guides **118** of the latch or stop member **40**, with the projections or fingers **120** and corresponding cam members **126** of the latch or stop member **40** positioned in the grooves or guides **78** of the indexer **38** wherein the cam surface or ramp **84** of the projection or finger **80** contacts the cam surface or ramp **138** of the projection or finger **92** of the translating member **46**. The force applied by the return members or springs **26** maintains engagement between the cam surface or ramp **84** of the projection or finger **80** of the indexer **38** and the cam surface or ramp **138** of the projection or finger **92** of the translating member **46**, wherein the end **142** of the projection or finger **92** of the translating member **46** contacts and rests on the end wall or surface **124** of the groove or guide **118** of the latch or stop member **40**.

(52) The cam surface or ramp **72** of the first or outer cam member **70** of the indexer **38** contacts the cam surface or ramp **96** of an adjacent first or outer cam member **94** of the translating member **46**. The respective cam surfaces or ramps **72**, **96** are shown contacting one another approximately midway along the respective cam surfaces or ramps **72**, **96**. Because the stator **42** is not engaged, the first or outer cam members **94** of the translating member **46** apply no force to the indexer **38**. Depending upon tolerances or design, there may be a slight gap or space between the respective cam surfaces or ramps **72**, **96** when the stator **42** is off. When the stator **42** is off, the translating

member **46**, particularly the projection or finger **92**, transfers the load from the spring plate carrier **36** through the indexer **38** to the latch or stop member **40**. When the stator **42** is actuated, the translating member **46** applies force to the indexer **38** to move the indexer **38** axially, in the direction of the arrow **150**, toward the pocket plate **14**, enabling it to, due to interaction with the translating member **46** and the latch or stop member **40**, rotate and position the coupling device **12** in one of the “on” or “off” positions.

(53) Actuation of the stator **42** pulls the translating member **46** and translator **44** combination toward the stator **42** using electromagnetics. Pulling the translating member **46** and translator **44** combination toward the stator **42** moves the translating member **46** and translator **44** combination axially, in the direction of the arrow **150**. The engagement configuration between the first or outer cam members **70** of the indexer **38** and the first or outer cam members **94** of the translating member **46** stays, as shown in FIG. 5, until the projections or fingers **80** and corresponding second or inner cam members **82**, including the cam surfaces or ramps **84**, pushed axially in the direction of the arrow **150** by the projections or fingers **92** of the translating member **46**, clear the grooves or guides **118** between adjacent projections or fingers **120** of the latch or stop member **40**.

(54) FIG. 7 shows, once the projections or fingers **80** of the indexer **38** clear the grooves or guides **118** of the latch or stop member **40** the force applied by the return member or spring **26** acts, through the spring plate carrier **36**, on the indexer **38** and correspondingly the adjacent cam surfaces or ramps **72**, **96** of the indexer **38** and the translating member **46**. The force, through the cam surface or ramps **72**, **96**, of the first or outer cam member **70** of the indexer **38** and adjacent first or outer cam member **94** of the translating member **46**, rotates the indexer **38** in the direction shown by arrow **152**. As the indexer **38** rotates to the position in FIG. 7, the cam surface or ramp **72** of the first or outer cam member **70** of the indexer **38** slides along the respective cam surface or ramp **96** of the first or outer cam member **94** of the translating member **46** until the cam displacement surfaces **74**, **98** of the outer cam members **70**, **94** engage or contact one another. Engagement of cam displacement surfaces **74**, **98** stops rotation, movement in the direction of the arrow **152**, of the indexer **38**. The amount or degree of rotation is determined by the circumferential gap or space **154** between the apex **156** of first or outer cam member **70** of the indexer **38** and the vertex **158** of first or outer cam member **94** of the translating member **46**. The indexer **38** rotates due to the axial force applied by the stator **42** to the translator **44** and translating member **46** acting against the return member or spring **26** and on the cam surfaces or ramps **72**, **96**. The cam surfaces or ramps **84**, **138** of the second or inner cam members **82**, **136** form complementary ramp surfaces on which the cam members **82**, **136** slide. FIG. 7 shows the rotation of the indexer **38** stopped with the projection or finger **80** and corresponding cam surface or ramp **84** of the second or inner cam member **82** positioned slightly over, or above, the cam surface or ramp **138** of the projection or finger **92** of the second or inner cam member **136** of the translating member **46**.

(55) De-energizing the stator **42** removes the axial force applied by translator **44** on the translating member **46**. FIG. 6 shows moving the translator **44** and the translating member **46** in the direction shown by the arrow **160**, disengages the indexer **38** and translating member **46**, wherein the outer cams **70**, **94** separate from one another. The force provided by the return member or spring **26** slides the cam surface or ramp **84** of the second or inner cam **82** of the indexer **38** along the angled cam surface or ramp **128** of the cam member **126**, rotating the indexer **38** in the direction of the arrow **152**. The translating size, axial amount, or degree of the circumferential gap or space **154** between the indexer **38** and translating member **46** is shown for illustrative purposes only and may vary. The respective cam displacement surfaces **74**, **85** are angled or inclined with respect to the longitudinal axis **34**. As the indexer **38** rotates and moves, in the direction of the arrow **152**, the angled or inclined cam displacement surface **74** acts on the corresponding angled or inclined cam displacement surface **98** to help move and allow movement of the translating member **46** in the direction of the arrow **160**, away from the indexer **38**.

(56) The return member or spring **26** also moves the indexer **38** axially in the direction shown by

arrow **160**, wherein the cam surface or ramp **84** of the projection or finger **80** engages a corresponding angled cam surface or ramp **128** of the cam member **126** of the latch or stop member **40**. Engagement of the cam surface or ramp **84** and the corresponding angled cam surface or ramp **128** rotates the indexer **38** in the direction of the arrow **152**. Once again, the indexer **38** rotates in the direction of the arrow **152** due to the axial force applied by the return member or spring **26** and the angled cam surfaces or ramps **84**, **128**. Sliding or relative movement between the adjacent cam surfaces or ramps **84**, **128** rotates the indexer **38** about the longitudinal axis **34** in the direction of arrow **152**. As shown in FIG. **8**, the projection or finger **80** of the indexer **38** now rests in the shallow notch **144** between adjacent cam surfaces or ramps **128** and cam displacement surfaces **130** placing the clutch in the “on” position.

(57) Moving the clutch from the “on” position to the “off” position requires re-energizing the stator **42**. Upon energizing the stator **42**, the translator **44** and translating member **46** combination moves axially in the direction of arrow **150**. The cam surfaces or ramps **72**, **96** again engage one another. The translating member **46** acts against the return member or spring **26**, moves the indexer **38** axially, and lifts the corresponding projections or fingers **80** of the indexer **38** out of the notch **144** formed between adjacent cam surfaces or ramps **128**. Once the projections or fingers **80** are clear of the notch **144**, the indexer **38** rotates in the direction of the arrow **152** until the cam displacement surface **85** of the second or inner cam member **82** of the translating member **46** engages the cam displacement surface **140** of the second or inner cam member **136** of the translating member **46**, the respective sides of adjacent projections or fingers **80**, **92** engage one another; see FIG. **9**. In addition, the respective cam displacement surfaces **74**, **98** engage one another. The cam surface or ramp **84** is positioned over the cam surface or ramp **128** adjacent to the groove or guide **118** in the latch or stop member **40**. FIG. **9** shows the indexer **38** rotating approximately 3-5° to align the adjacent cam surfaces or ramps **84**, **128**. When the stator **42** is de-energized, the return member or spring **26** moves the indexer **38** in the direction of the arrow **160**, wherein the cam surface or ramp **84** slides on the cam surface or ramp **128** and rotates the indexer **38** in the direction of the arrow **152**. The rotation causes inclined or angled cam displacement surface **74** to slide up and over correspondingly inclined or angled cam displacement surface **98**. Indexer **38** rotation continues until the projections or fingers **80** of the indexer **38** enter and slide in the grooves or guides **118**—locating the projections or fingers **80** in the grooves/guides **118** places the actuation mechanism **10** and correspondingly the clutch in the “off” position, as illustrated in FIG. **6**.

(58) The rotation or amount of rotation of the indexer **38** is determined by the angle of the cam surfaces **72**, **74**, **84**, **85**, **96**, **98**, **138**, **140**, **128**, and **130**. Continued rotation of the indexer **38** continues the extend/retract cycle by repeating the same steps.

(59) The foregoing disclosure incorporates using a plurality of different cams and indexers rotating about a longitudinal axis of a clutch shaft.

(60) The foregoing provides an actuation mechanism requiring zero power while in state and in one direction of actuation regardless of traveling from “on” to “off” or “off” to “on.” The foregoing works well in situations where the parts rotate-centrifugally neutral.

(61) FIGS. **10-15** illustrates alternative examples of a mechanism for moving the translating member **46**. For example, rather than using a stator **42**, a double-sided shift fork, generally seen at **170**-FIGS. **10-11**; a single-sided shift fork, generally seen at **190**-FIGS. **12-13**; and a pneumatic piston, generally seen at **210**-FIGS. **14-15** may also move the translating member **46**. The foregoing are illustrative examples and do not limit the mechanisms that move the translating member **46**.

(62) FIGS. **10** and **11** show the double-sided shift fork **170** includes a C-shaped portion **172**, including two radially inwardly extending flanges **174**, **176** forming a notch **178**. A radially outwardly extending flange **180** on the translating member **46** is in the notch **178**. The double-sided shift fork **170** also includes a connection portion **182** having an aperture or bore **184** sized to receive a member (not shown) that moves the shift fork **170** reciprocally in the direction of the

longitudinal axis **34**. The double-sided shift fork **170** moves the translating member **46** between the corresponding “on” and “off” positions. Moving the translating member **46** rearwardly, toward the end **104** of the latch or stop member **40** in the direction of the arrow **160** pulls the translating member **46** rearwardly and disengages the respective cam members **70**, **94** of the indexer **38** and translating member **46**, whereby the return member or spring **26** moves the indexer **38** rearward in the direction of the arrow **160**. The cam surface or ramp **72** slides on the cam surface or ramp **96**, rotating the indexer **38** in the direction of the arrow **152**. As shown, the inwardly extending flange **176** acts on the flange **180** of the translator **44** connected to the translating member **46** to pull the translating member rearward.

(63) FIGS. **12** and **13** illustrate an embodiment similar to that in FIGS. **10-11**, except the shift fork, is a single-sided shift fork **190**. The shift fork **190** includes a single inwardly extending flange member **192** on a C-shaped portion **194**. As shown, the flange member **192** is positioned on one side of the flange **200** of the translator **44**, connected to the translating member **46**, on the side closest to the end **104** of the latch or stop member **40**. The double-sided shift fork **190** also includes a connection portion **196** having an aperture or bore **198** sized to receive a member (not shown) that moves the shift fork **190** reciprocally in the direction of the longitudinal axis **34**. In this embodiment, the single-sided shift fork **190** operates like the embodiment with the stator **42**. The shift fork **190** pushes the translator **44** and translating member **46** axially in the direction of the arrow **150**, while the return member or spring **26** moves the translating member **46** axially in the direction of the arrow **160**.

(64) FIGS. **14** and **15** illustrate a piston **210** used to move the translator **44** and translating member **46**. The piston **210** includes a housing **212** and piston **214**. The piston **214** is connected to the translator **44**. A pair of seal members, illustrated as O-rings **216**, in grooves **218**, form a seal between the housing **212** and the piston **214**. Fluid flowing through the aperture **220** acts on the piston **214**, which moves the translator **44** and translating member **46**. Like the embodiment with the stator **42**, the piston **210** moves the translating member **46** axially in the direction of the arrow **150**, while the return member or spring **26** moves the translating member **46** axially in the direction of the arrow **160**.

(65) The foregoing mechanisms can be used separately or in combination. For example, the actuation mechanism may include a stator **42** and other mechanisms to move the translating member **46** back and forth. In addition, a multiple-position linear actuator may also control the movement of the translating member **46** in both directions.

(66) FIG. **16** illustrates the stator **42** and translator **44** of the actuation mechanism **10**. The stator **42** is stationary and is fixed to the stationary member, for example a transmission case (not shown). The stator **42** having a housing **220** and an electromagnetically inductive coil **222** positioned between adjacent ferromagnetic fingers **224**. The electromagnetically inductive coil **222** creates a magnetic flux when energized.

(67) The translator **44**, coupled to the translating member **46**, is mounted for rotational and axial movement on the hub **32**. The translator **44** is radially spaced from the stator **42**, with a radial gap **226** and an axial gap **228** between the stator **42** and translator **44**. The gaps **226**, **228** are sized such that the translator **44** is spaced from and does not contact the stator **42**. Both gaps **226**, **228** provided for tolerance differences between the respective stator and translator while maintaining a spacing between them. In one example the gap is in the order of point-blank millimeters. The smaller the gap, the less current needed to move the translator **44** and corresponding translating member **46**.

(68) The magnetic flux lines **230**, flow along closed paths with the majority of lines extending in a circular path across the radial gap **226** between the stator **42** and translator **44**. The flow of the flux lines produced by the stator **42** result in a net force on the translator **44**. The magnetic flux lines cross the gap **226**, **228** wherein the electromagnetically inductive coil **222** and the coil current drives the translator **44** in the direction of the arrow **150**.

(69) The electromagnet source or electromagnetically inductive coil **222** creates an electronically switched magnetic field across the radial and axial gaps **226**, **228** that acts on the translator **44** causing translational movement of the translator **44** and corresponding translating member **46** relative to the stator **42** along the longitudinal axis **34** in the direction of the arrow **150**.

(70) The stator **42** provides a pulse actuator, wherein energizing, supplying an electrical current to, the electromagnetically inductive coil **222** creates the magnetic field and moves the translator **44**. Deenergizing, removing or turning off the electrical current to, the electromagnetically inductive coil **222** removes the magnetic field and correspondingly allows the return spring **26** to act on the indexer **38** and corresponding translator hub **46** moving them rearwardly, in the direction of the arrow **160**.

(71) The stator **42** operates as a pulse actuator, the electrical current supplied to the electromagnetically inductive coil **222** may be supplied as a pulse, the duration and degree of which supplied in an amount to move the translator **44**, translating hub **46** and indexer **38** to a position wherein the fingers **80** of the indexer **38** clear the grooves **118** of the latch hub **40** as shown in FIG. 7. Once the fingers **80** are clear, the electric current to the electromagnetically inductive coil **222** turned off wherein the return springs **26** act on the indexer **38** moving both the indexer and the translating hub **46** in the direction of the arrow **160**.

(72) The stator **42** and translator **44** cooperate to move, pull the translating hub **46** to the right, in the direction of the arrow **150**, to both mechanically latch and mechanically unlatch the actuation mechanism **10**, and place the coupling device **12** in a deployed and non-deployed position.

(73) FIG. 17 illustrates an alternative embodiment of an actuation mechanism **10** having a different stator and translator assembly. Instead of the translator being an annular member, as in the first example, the translator **44** is a radial ring **162** attached to and extending radially outward from the translating member **46**.

(74) FIG. 18 illustrates another example of an actuation mechanism **10** wherein the cam displacement surface **164** of the indexer **38** and cam displacement surface **166** of the translating member **46** extend axially, parallel to the longitudinal axis **34** of the hub **32**. Because the cam displacement surfaces **164**, **166** are not angled but extend parallel to the longitudinal axis **34** of the hub **32**, engagement of the respective cam displacement surfaces **164**, **166** prevents rotation of the indexer **38**. This example requires an actuator that exerts a force that pulls the translating member **46** rearward or back, in the direction of the arrow **160**, to disengage the translating member **46** and indexer **38**. For example, the double-sided shift fork **170** would operate to both push and pull the translator **44** and translating member **46**.

(75) The actuation mechanism **10** provides a two-position latch, a first, retracted position, and a second, extended position. In the first, retracted position the second or inner set of cam members **82** of the indexer **38** are in grooves or guides **118** of the latch or stop member **40**. In one example, the second or inner set of cam members **82** engage a second or inner set of cam members **136** of the translating member **46**. In the second position, the second or inner set of cam members **82** engage the cam members **126** of the latch or stop member **40**. In the first and second positions, the latch or stop member **40** provides positive support; i.e., force applied to the indexer **38** in the direction of the arrow **160** is supported by the latch or stop member **40**. The latch or stop member **40** provides a fixed, hard stop or hard lock limiting axial movement of the indexer **38**.

(76) While examples or exemplary embodiments are described above, it is not intended these embodiments describe all possible forms of the present invention. The words used in the specification are words of description rather than limitation. It is understood that various changes may be made without departing from the spirit and scope of the present invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the present invention.

(77) The description of the invention is merely exemplary; thus, variations that do not depart from

the gist of the invention are intended to be within the scope of the invention. Such variations are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. An assembly comprising: a hub, the hub including a shaft; a first member secured to the hub and constrained against rotational and axial movement with respect to the hub, the first member having a side surface extending between an inner and an outer circumferential surface of the first member, the side surface of the first member including a cam surface; a second member rotatably fixed and axially movable with respect to the first member, the second member having a side surface extending between an inner and an outer circumferential surface of the second member, the side surface of the second member including a cam surface; a third member rotatably and axially movable with respect to the shaft, the third member having a side surface extending between an inner circumferential surface and an outer circumferential surface, the side surface of the third member including a first cam surface and a second cam surface extending at a same angle, the second cam surface of the third member radially inward of the first cam surface of the third member, the third member having a groove in the inner circumferential surface of the third member; and the cam surface of the first member is in the groove of the third member when the assembly is placed in a retracted position, and the cam surface of the first member contacts the second cam surface of the third member when the assembly is placed in an extended position.
2. The assembly of claim 1 including: the second cam surface of the third member is in a groove of the first member when the assembly is placed in the retracted position.
3. The assembly of claim 1 including: the third member engages a drive member; and the drive member engages a locking element.
4. The assembly of claim 1 including: the third member is supported by the first member in both the retracted position and the extended position.
5. The assembly of claim 1 including: a coupling member having a face and a at least one pocket communicating with the face; a locking element disposed in the pocket; and the locking element extends from the pocket and above the face when the locking element is in a deployed position, and the locking element remains within the pocket and below the face when the locking element is in a non-deployed position.
6. The assembly of claim 1 wherein: the first member is constrained against rotational and axial movement with respect to the hub.
7. The assembly of claim 1 including: a translator carried on the second member.
8. The assembly of claim 1 including: a carrier carried on the hub, the carrier including a cylindrical body and a radially outwardly extending flange; and the third member carried on the cylindrical body of the carrier.
9. The assembly of claim 1 wherein: the first cam surface and the second cam surface of the third member extending to an end of the third member.
10. The assembly of claim 1 including: the second member having a second cam face, the second cam face engaging the second cam face of the third member.
11. An assembly comprising: a first member having a cylindrically shaped body, a cam face and an end face opposite the cam face, the first member including a plurality of projections and grooves on an outer circumferential surface of the first member; a second member having a cylindrically shaped body, a cam face and a plurality of projections and grooves on an inner circumferential surface of the second member, the projections and grooves on the inner circumferential surface of the second member complementary to the projections and grooves on the outer circumferential surface of the first member wherein engagement of the projections and grooves on the outer circumferential surface of the first member with the projections and grooves on the inner circumferential surface of the second member allows axial movement and prevents rotational

movement between the first member in the second member; the cam face of the second member spaced radially from and concentric with the cam face of the first member; a third member having a cylindrically shaped body, a cam face and a plurality of projections and grooves on an inner circumferential surface of the third member, the cam face of the third member including a first cam face and a second cam face, the first and second cam faces have a common border; and the first cam face of the third member engages the cam face of the second member, and the second cam face of the third member engages the cam face of the first member.

12. The assembly of claim 11 including: a coupling member having a face and at least one pocket communicating with the face; a locking element disposed in the pocket; a carrier, the carrier engaging the third member, the carrier is rotatably fixed and axially movable for axial motion with no rotation; and a drive member, the drive member between the carrier and the locking element wherein the locking element extends from the pocket and above the face when the locking element is in a deployed position, and the locking element remains within the pocket and below the face when the locking element is in a non-deployed position.

13. The assembly of claim 11 including: the second member having a second cam face, the second cam face engaging the second cam face of the third member.

14. The assembly of claim 11 including: the first cam face and the second cam face of the third member extending to an end of the third member.

15. An assembly comprising: a hub, the hub including a shaft rotating about a longitudinal axis; a first member secured to the hub and constrained against rotational and axial movement with respect to the hub, the first member having a side surface extending between an inner and an outer circumferential surface of the first member, the side surface of the first member including a cam member; a second member rotatably fixed and axially movable with respect to the first member, the second member having a side surface extending between an inner and an outer circumferential surface of the second member, the side surface of the second member including a cam member; a carrier carried on the hub, the carrier having a cylindrical body and a radially outwardly extending flange; a third member positioned over the cylindrical body and rotating around the cylindrical body, the third member rotatably and axially movable with respect to the shaft, the third member having a side surface extending between an inner circumferential surface and an outer circumferential surface, the side surface of the third member including a first cam member and a second cam member, the second cam member of the third member radially inward of the first cam member of the third member, and the third member having a groove in the inner circumferential surface of the third member; and the cam member of the first member is in the groove of the third member when the assembly is placed in a retracted position, and the cam member of the first member contacts the second cam member of the third member when the assembly is placed in an extended position.

16. The assembly of claim 15 including: the second cam member of the third member having a cam ramp sloped with respect to the longitudinal axis of the hub; and a cam displacement surface extending parallel to the longitudinal axis.

17. The assembly of claim 15 including: the second member having a second cam member, the second cam member of the second member engaging the second cam member of the third member.

18. The assembly of claim 17 including: the second cam member of the second member having a cam ramp that extends at the same angle and in common with a cam ramp of the first cam member of the second cam member.

19. The assembly of claim 15 including: the second cam member of the third member having a cam ramp that extends at the same angle and in common with a cam ramp of the first cam member of the third member.
